

Corrigendum to “Micro-foundations of Congestion and Pricing: A Game Theory Perspective”

David Levinson*

Nicholas Janusch†

May 26, 2015

Abstract

This paper amends “Micro-foundations of Congestion and Pricing: A Game Theory Perspective” Levinson [2005], correcting the calculated tolls and Nash equilibria predicted for a three-player game.

1 Introduction

In reviewing and extending “Micro-foundations of Congestion and Pricing: A Game Theory Perspective” Levinson [2005], Nicholas Janusch identified several errors in the original manuscript. The following are corrections to Levinson [2005]:

- p700: Probabilities in Table 10 for Player A departing Late (l) given Players B and C departing Early and Late (el) are incorrect. $P(\bar{r}) = 0.5$ not 0 for this scenario.
- p700: In the sentence “To illustrate the gains from pricing, for instance, in the case: $E = 3$, $D = 1$, $L = 4$, the unpriced equilibrium solution eee has a total cost of 10, compared with the priced equilibria of eeo , oeo , ooe which have a total cost of 7.” The total cost is 8 not 7 ($3 + 0 + 1 + 4 = 8$).
- p701 Corrections to several predicted pure strategy Nash equilibria (PSNE) reported in Table 11 using this paper’s three-player game methodology. ($\#, \#, \# = E, D, L$). The revised Table 11 is shown.

*RP Braun-CTS Chair of Transportation Engineering; Director of Network, Economics, and Urban Systems Research Group; University of Minnesota, Department of Civil, Environmental, and Geo- Engineering, 500 Pillsbury Drive SE, Minneapolis, MN 55455 USA, dlevinson@umn.edu <http://nexus.umn.edu>

†Department of Economics, St. Lawrence University, 23 Romoda Drive, Canton, NY 13617 Department of Agricultural and Resource Economics, Colorado State University, Fort Collins, CO 80523 njanusch@stlawu.edu, nicholas.janusch@colostate.edu

Table 1: Corrected Table 11 from Levinson (2005) of results from three-player game

E, D, L	Number of Nash equilibria (unpriced)	Solutions (Unpriced)	Number of Nash equilibria (priced)	Solutions (priced)	Lowest total cost
0, 0, 0	64	all	64	all	0
0, 1, 0	24	<i>veo, ..., vel, ..., vol, ..., eol, ...</i>	24	<i>veo, ..., vel, ..., vol, ..., eol, ...</i>	0
0, 0, 1	16	<i>vvv, vve, ..., vvo, ..., vee, ..., veo, ...</i>	16	<i>vvv, vve, ..., vvo, ..., vee, ..., veo, ...</i>	0
0, 1, 1	6	<i>veo, ...</i>	6	<i>veo, ...</i>	0
1, 0, 0	8	<i>ooo, lll, ool, ... , oll, ...</i>	8	<i>ooo, lll, ool, ..., oll, ...</i>	0
1, 1, 0	7	<i>ooo, ool, ..., oll, ...</i>	9	<i>oll, ..., eol, ...</i>	1
1, 0, 1	5	<i>ooo, eee, eoo, ...</i>	10	<i>eee, eeo, ..., eol, ...</i>	2
1, 1, 1	9	<i>eol, ..., eoo, ...</i>	6	<i>eol, ...</i>	2
3, 1, 4	1	<i>eee</i>	6	<i>eol, ...</i>	7
4, 0, 3	2	<i>ooo, eee</i>	10	<i>eee, eoo, ..., eol, ...</i>	7
4, 1, 3	4	<i>ooo, eoo, ...</i>	6	<i>eol, ...</i>	7
3, 0, 4	1	<i>eee</i>	13	<i>eee, eeo, ..., eoo, ..., eol, ...</i>	7

Note: Column 1 gives the penalty for arriving early, suffering journey delay, or arriving late. The characterization of solutions identifies which patterns are Nash equilibria, *v* indicates vehicles arrive very early, *e* early, *o* on-time, *l* late.

The original author regrets the error.

References

David Levinson. Micro-foundations of congestion and pricing: A game theory perspective. *Transportation Research Part A: Policy and Practice*, 39(7):691–704, 2005.